



Teamwork

in IT Development Projects

About the presenters

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About the presenters

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Agenda:

1. Agile Development
2. Scrum
3. Kanban
4. Tools
5. GIT and Workflows
6. Hands On GIT
7. Questions



Manifesto for Agile Software Development

We are uncovering better ways of developing software by doing it and helping others do it.
Through this work we have come to value:

Individuals and interactions over processes and tools
Working software over comprehensive documentation
Customer collaboration over contract negotiation
Responding to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.



<https://agilemanifesto.org/>

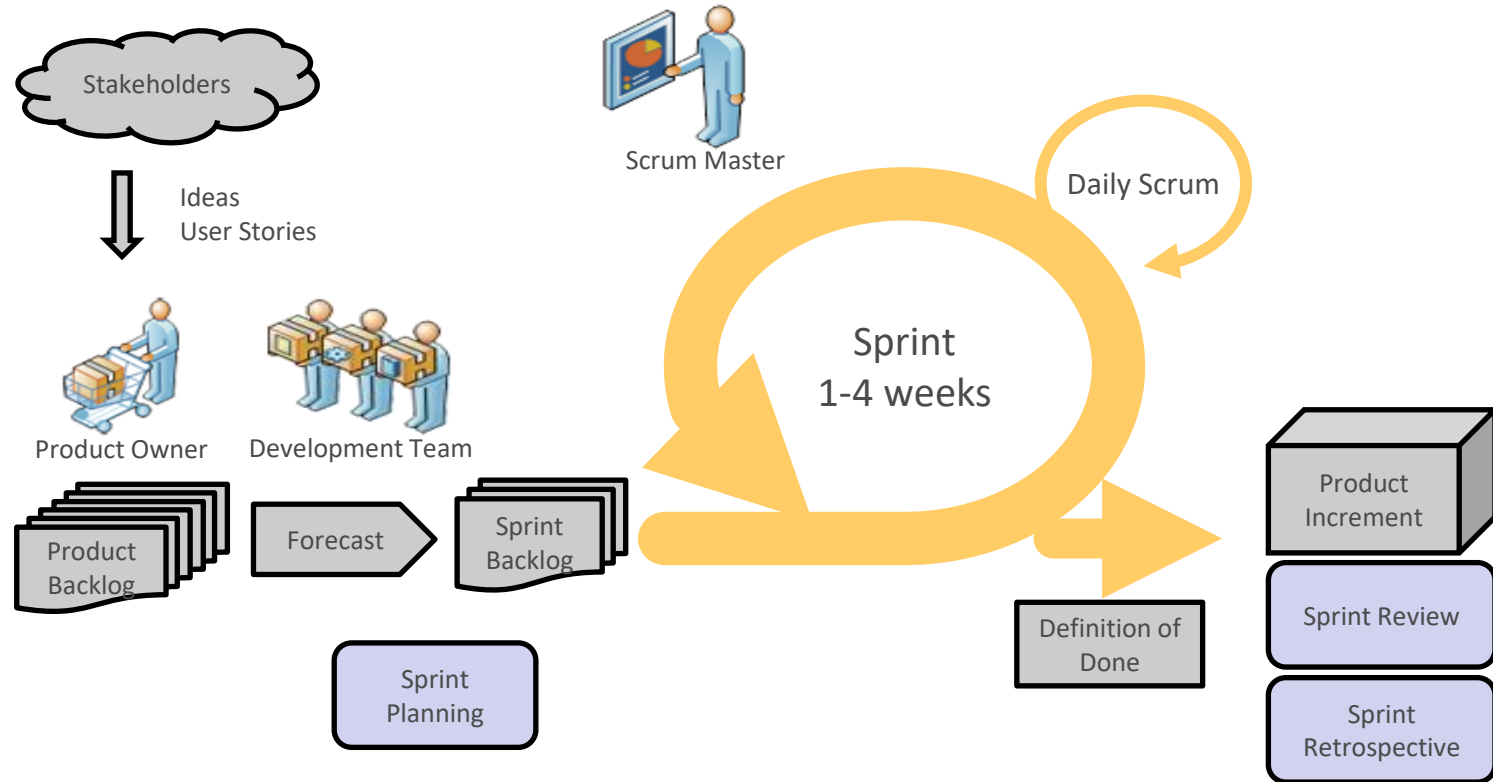


Scrum

How 90% of teams work in Cluj

Scrum

Defining Roles, Events and Artifacts



Scrum

Roles



Development Team

“Deliverer” of the Product.

Implements incremental parts of the product, for the Product Owner.

Plans sprints in personal responsibility



Product Owner (PO)

Represents the customer side.

Responsible to maximise the value of the product and the work of the development team

Defines and prioritises the requirements in the backlog.

This is a single person, not a committee. The organisation has to respect his decisions.



Scrum Master (SM)

Is responsible for the knowhow and execution of Scrum

Helps the team in their self-organisation and reflection.

Removes impediments in the team

Supports the PO managing the Backlog

Scrum

Events



Sprint

A Timebox (max 1 month), in which a product increment is produced.

All Sprints have the same length. The new Sprint starts immediately after finishing the last.

During a sprint a sprint goal and a quality expectations will not be adapted.



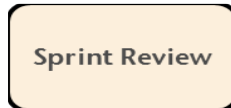
Sprint Planning

During the Sprint Planning the Scrum Team plans the new Sprint together. The Scrum Team is not only the development team!!!



Daily Scrum

Timebox of 15 min, coordination of the development team.



Sprint Review

Check of the product increment, adaptation of the backlog if needed. Goal is to get as much feedback as possible.



Sprint Retrospective

Überprüfung und Verbesserung der Zusammenarbeit

Scrum

Artifacts



Product Backlog

A ordered List of all things what should be part of the product.
Ist the only source of requirements for the whole application.
The Product Owner is responsible to manage the Backlog.



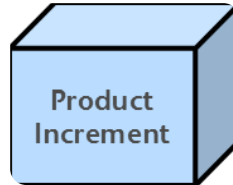
Sprint Backlog

The total amount of entries selected from the Product Backlog for the next sprint which will be created for the next product increment.
It's an estimation of the development team how much work they can finish.



Definition of Done

Defines what "Done« means, if a Backlog entry or a Product Increment is called «Done».
Increases the transparency through a clear definition, which needs to be followed by the team.

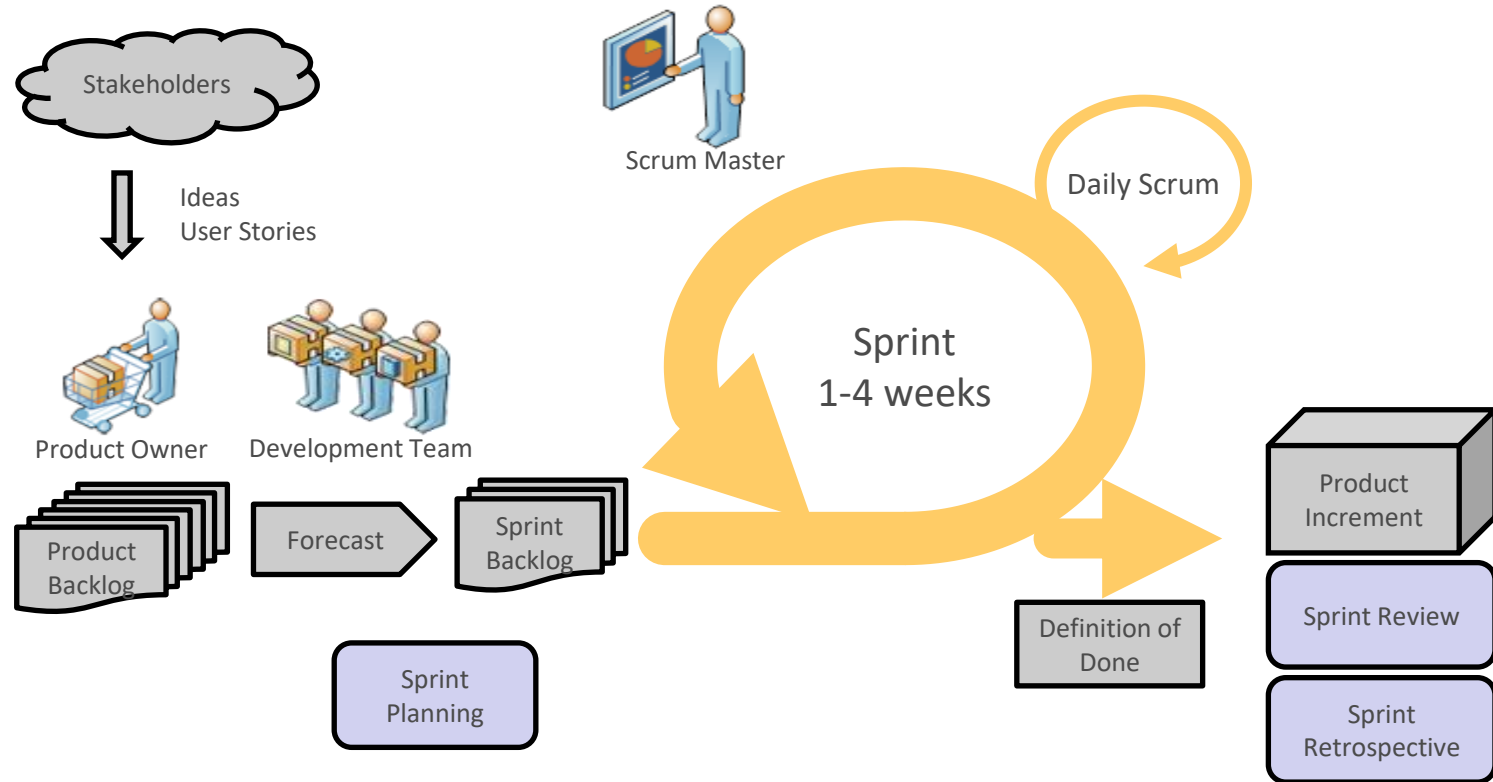


Product Increment

The result of all the entries which are done in the last sprint + all the results from previous sprints.

Scrum

Defining Roles, Events and Artifacts



Scrum

Artifacts



[User Story Template Design from Mike Cohen]

Scrum

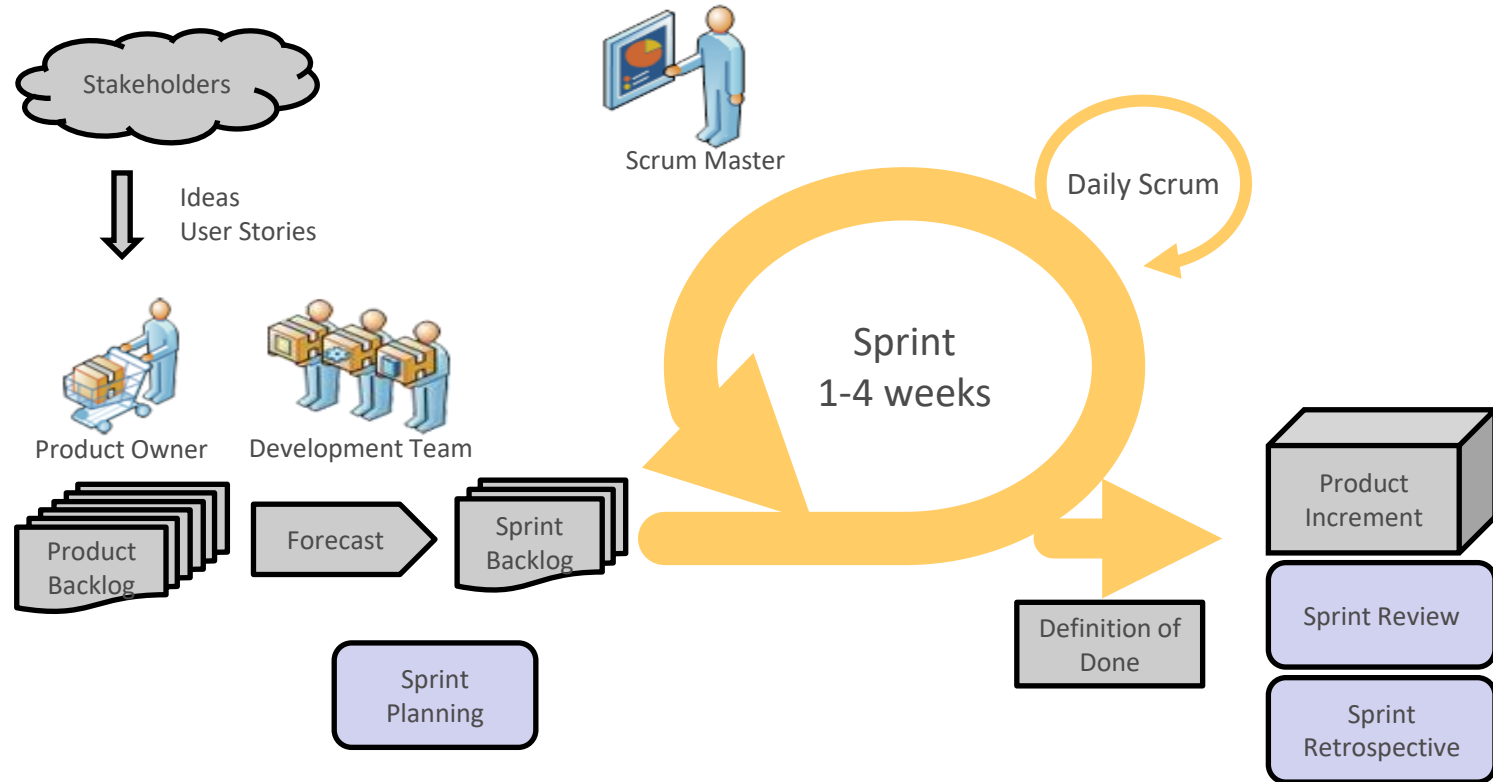
Artifacts



[User Story Template Design from Mike Cohen]

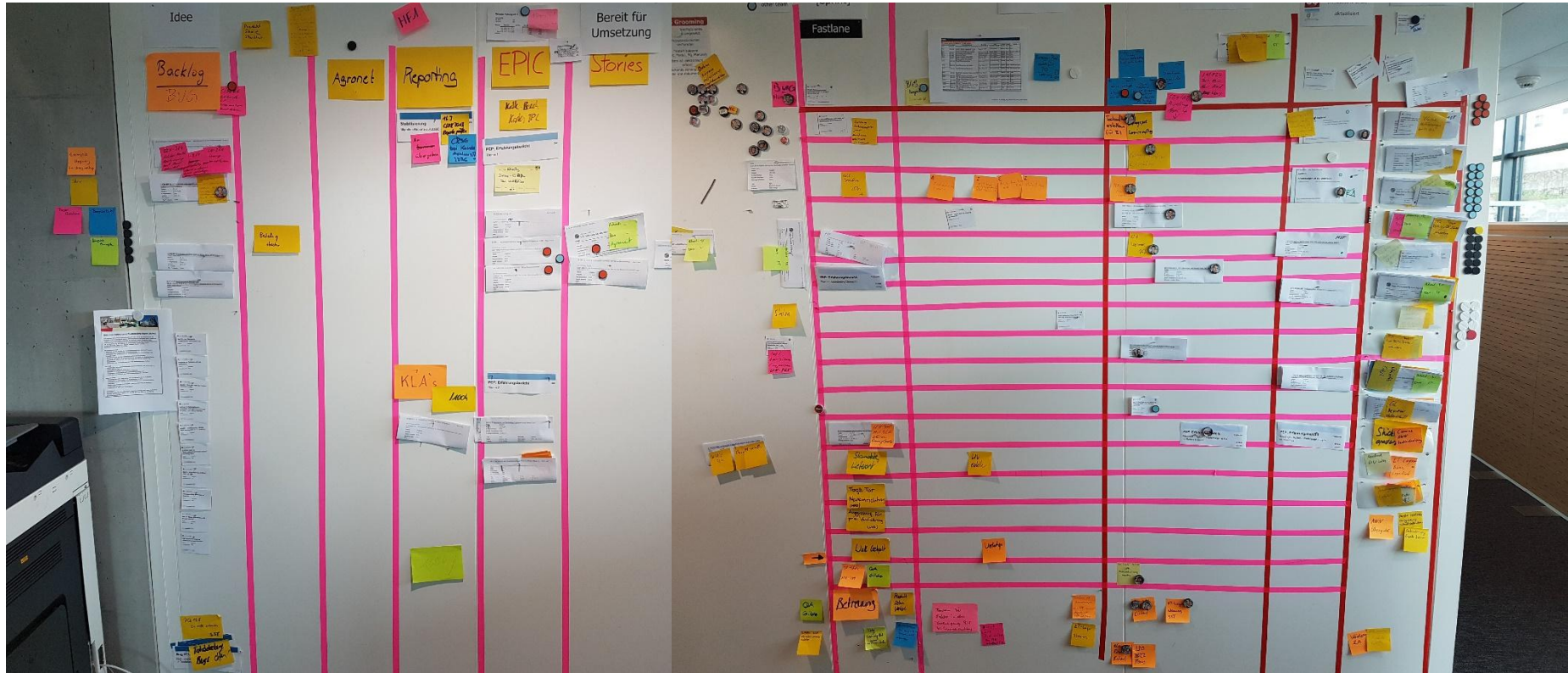
Scrum

Defining Roles, Events and Artifacts



Scrum

Examples





Kanban

Might be the better option for the student project

What is Kanban?

Creating a culture of continuous improvement

Kan

看

signal

Ban

板

card

Kanban overview

Pull principle with visual representation via cards

kanban

- Toyota Production systems, kanban is the core piece of visualisation of their production management
- Signal cards show downstream production steps, that a task is done, and that new material and components are needed
- Pull principle: reducing storage and visualize problems in the process

Kanban in IT

- Adapted system in software development
- Supports change evolutionary by optimizing processes successively
- Kanban in IT is based on lean (efficient design of a value chain)
- David Anderson is the creator of Kanban in IT
- Works especially well in Support and Maintenance

Kanban overview

Stop starting, start finishing

Kanban Base Principles

- Start with what you do already
- Pursue incremental, evolutionary change
- Respect the initial process, roles and responsibilities
- Support leadership in all levels of an organisation

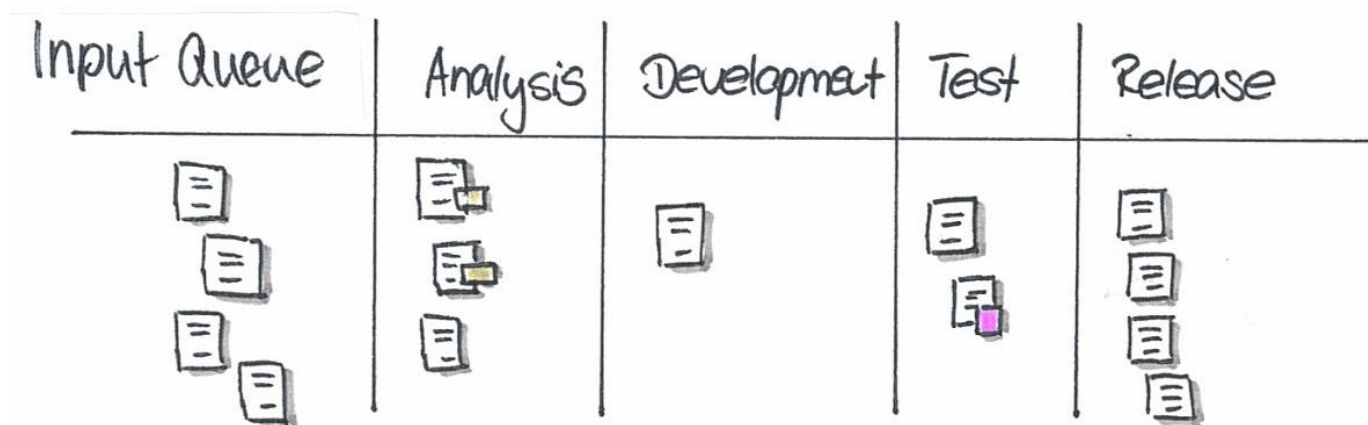
Kanban Core Practices

- Visualize work
- Limit the work in progress (WIP)
- Manage flow
- Make process steps explicit
- Implement fast feedback mechanisms
- Execute communal improvements

Visualize work

Visualize the value chain

Work will not be pushed into the next step. The workers from the next step pull the work once they have the necessary capacity. (Pull principle)

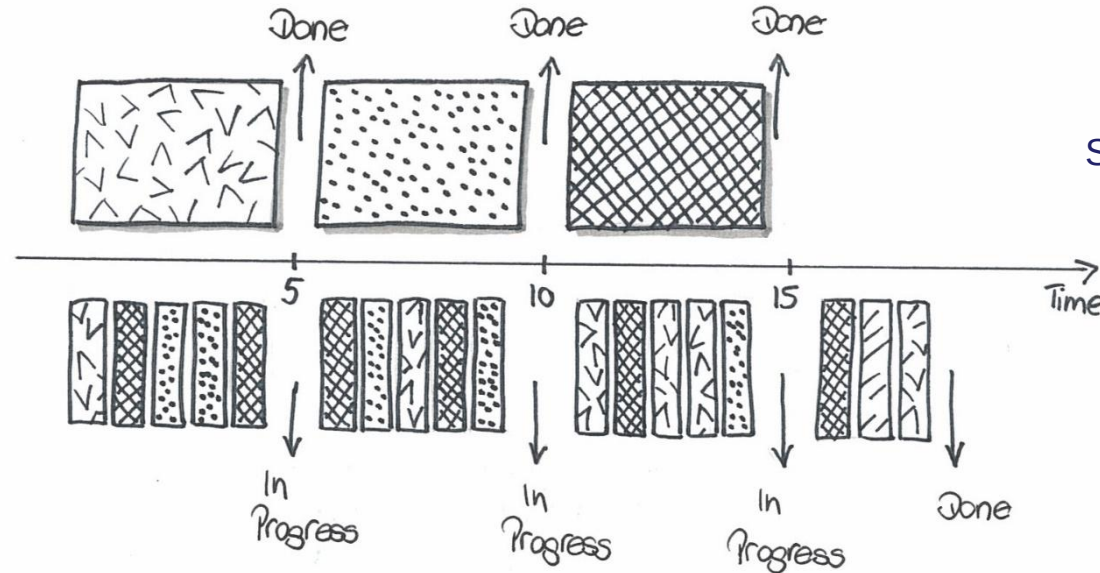


The Kanban Team identifies in which position of the value chain it's located, how the process works and defines the borders.

Limit the work in progress (WIP)

Difference between sequential and parallel work

A limit on the amount of work, which is executed in one step, reduces the time and improves the workflow.

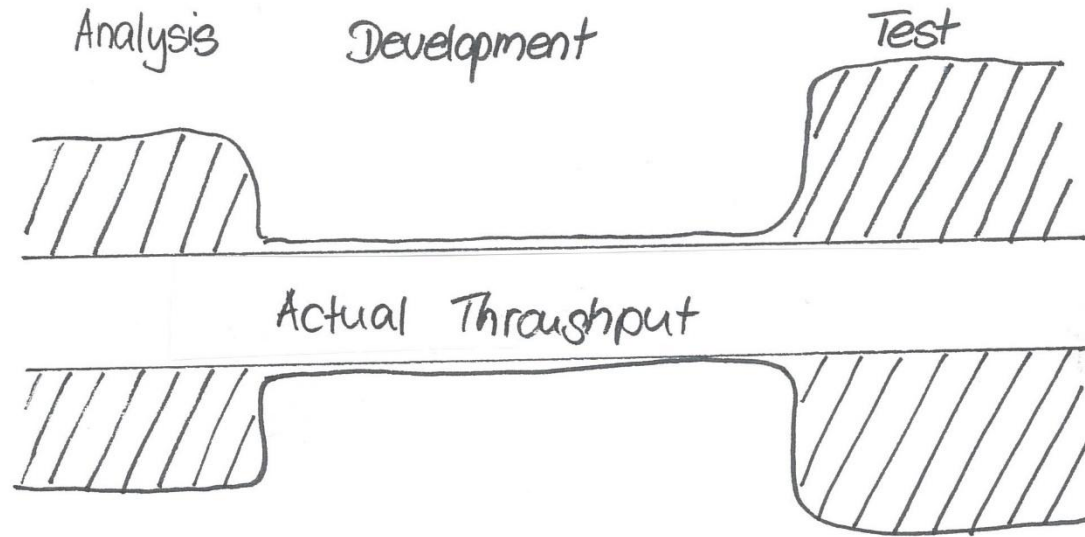


Sequential work → smaller worktime

parallel work → higher worktime

Manage Flow

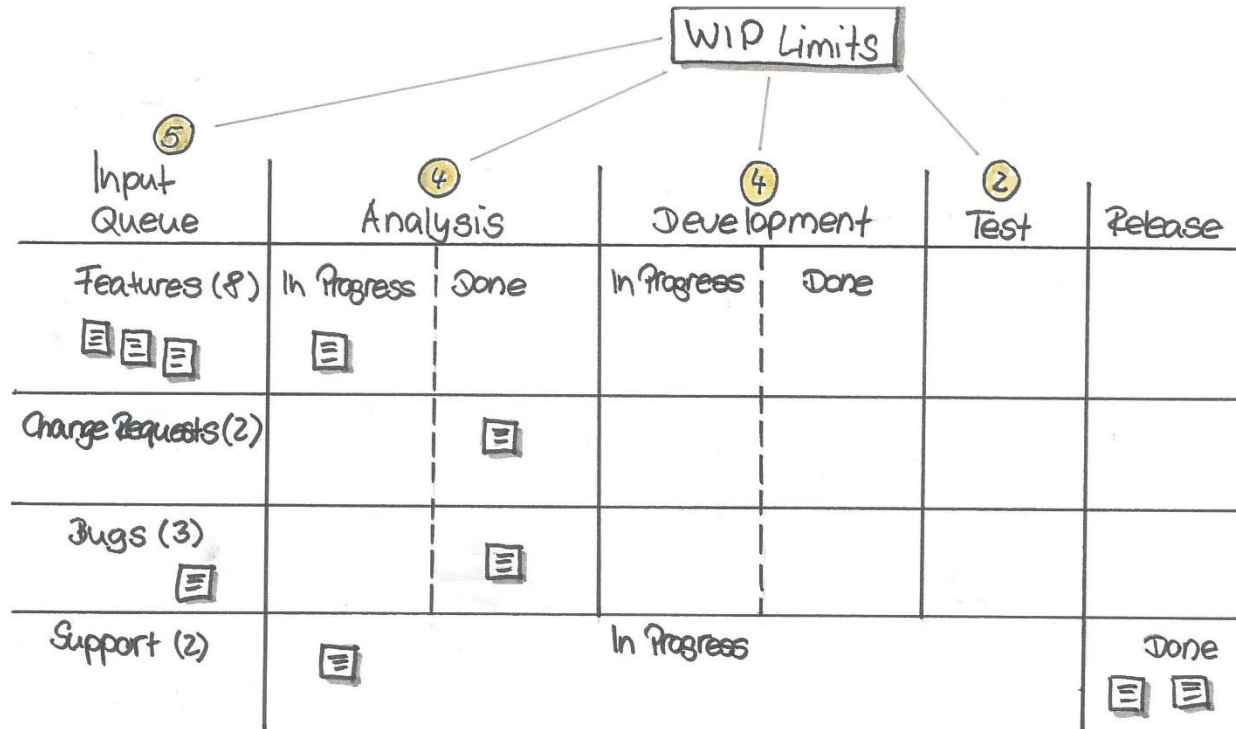
The total amount done is defined by the bottleneck of the system



Don't let more Tasks into the system than the bottleneck can handle. WIP limits are managed so that the bottleneck is not overwhelmed.

Manage Flow

Create a value chain with WIP limits per task type and process step



Questions



Meme-Generation

Thank You!

